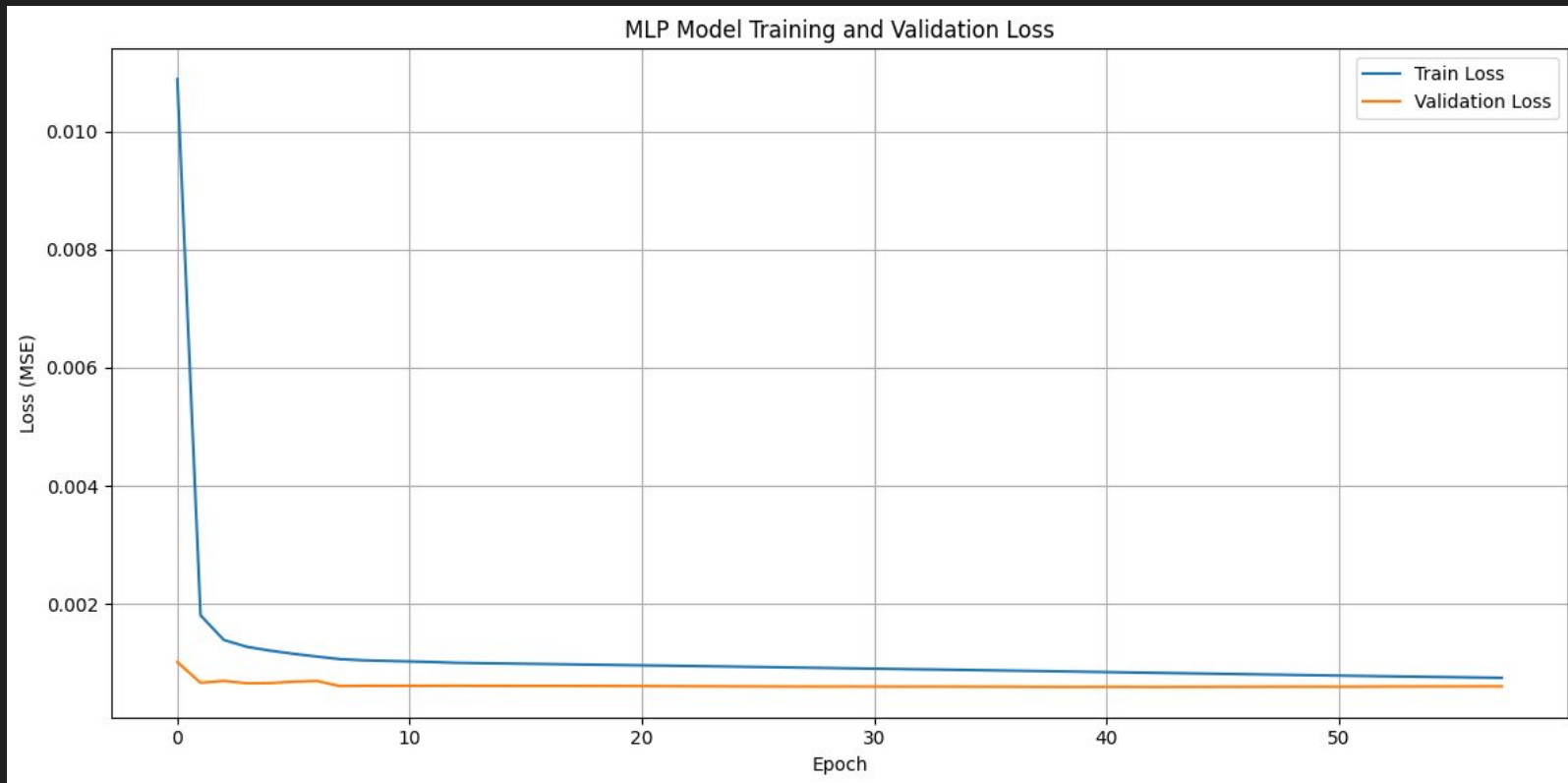
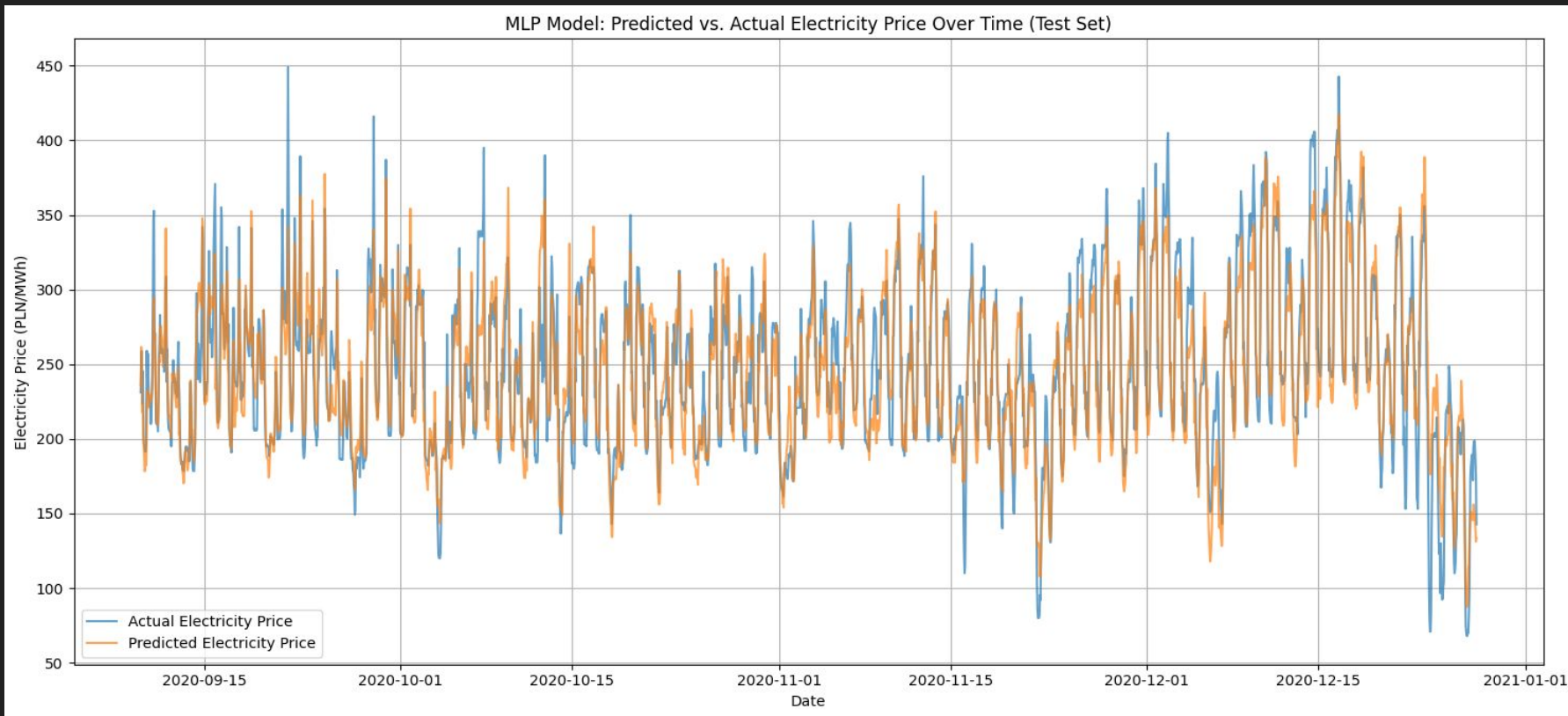


Presentation on short-term electricity price forecasting at the Polish Day-Ahead Market using machine learning models

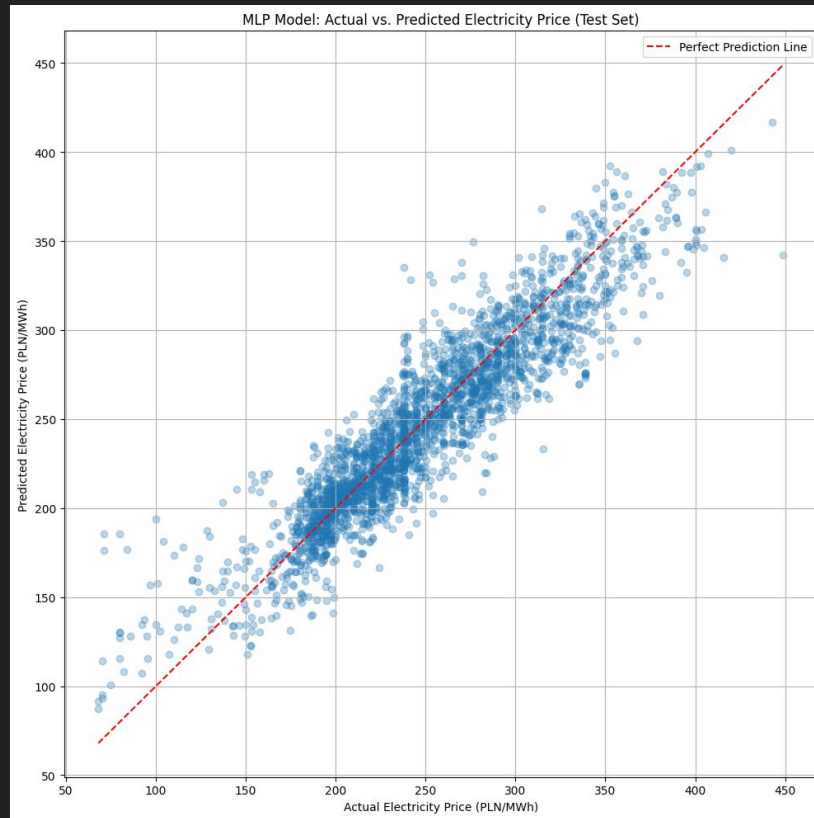
This plot shows stable convergence of the MLP model, with closely aligned training and validation loss, indicating good generalization and no evident overfitting



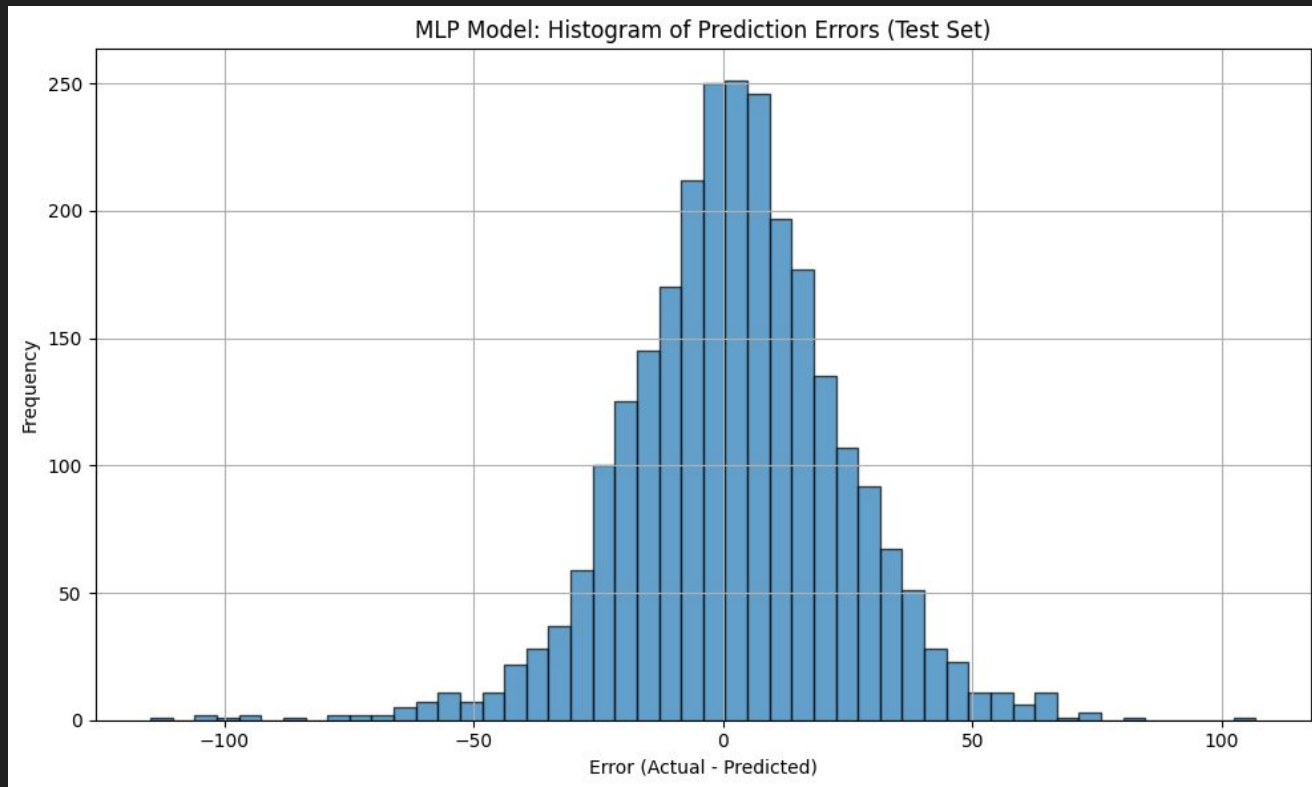
This plot shows that the MLP model closely follows the actual electricity price dynamics over time on the test set, capturing both short term fluctuations and overall trends



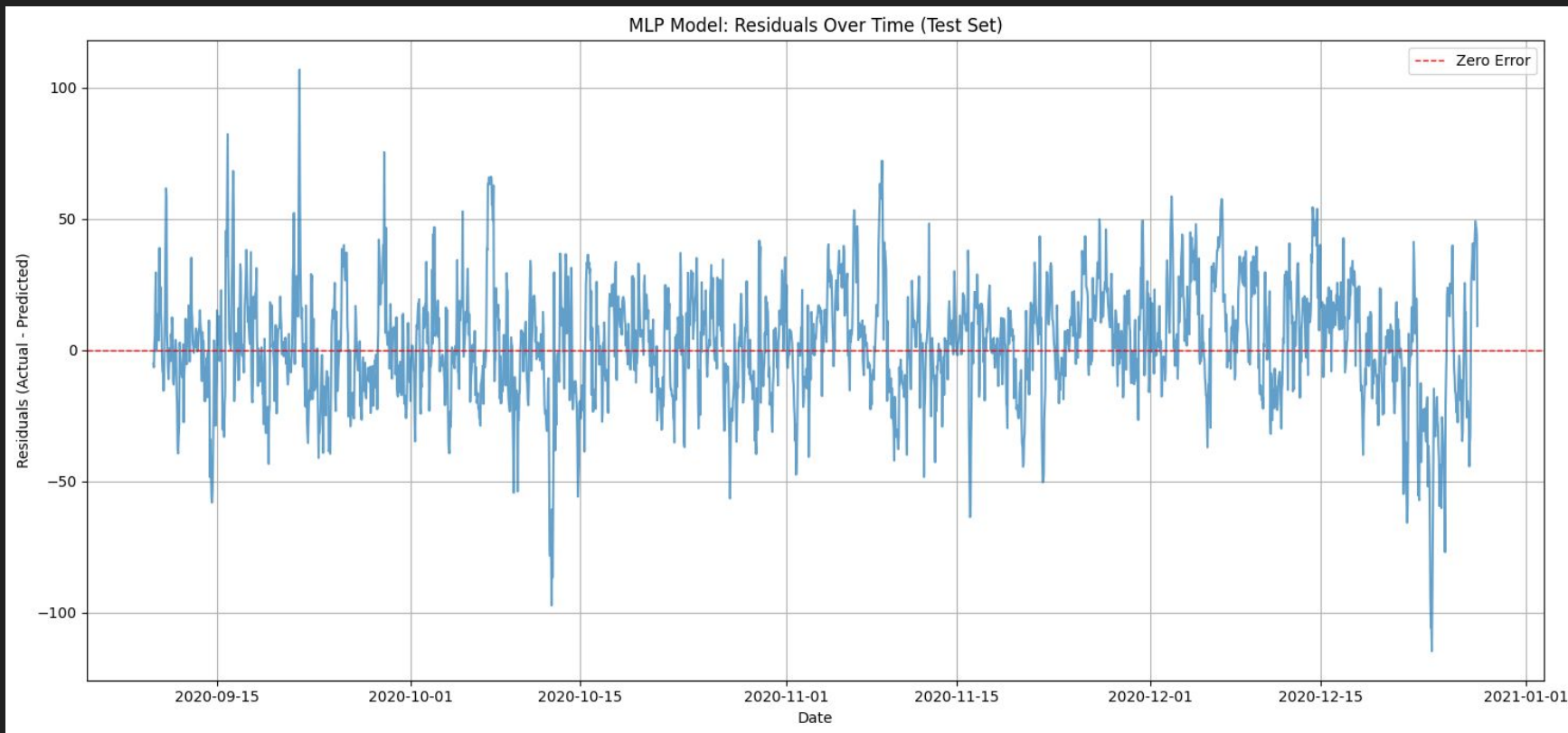
This scatter plot shows a strong linear relationship between actual and predicted electricity prices on the test set, indicating high prediction accuracy with moderate dispersion at extreme values



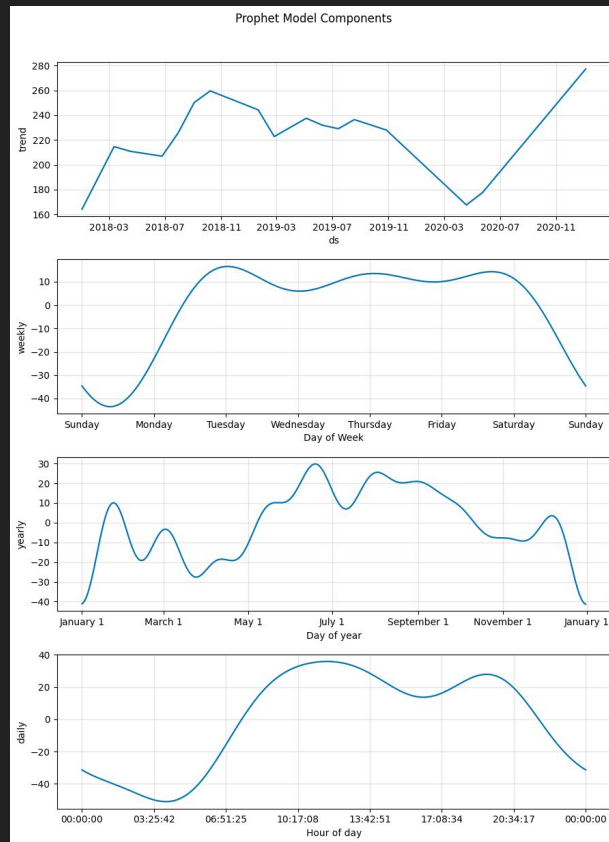
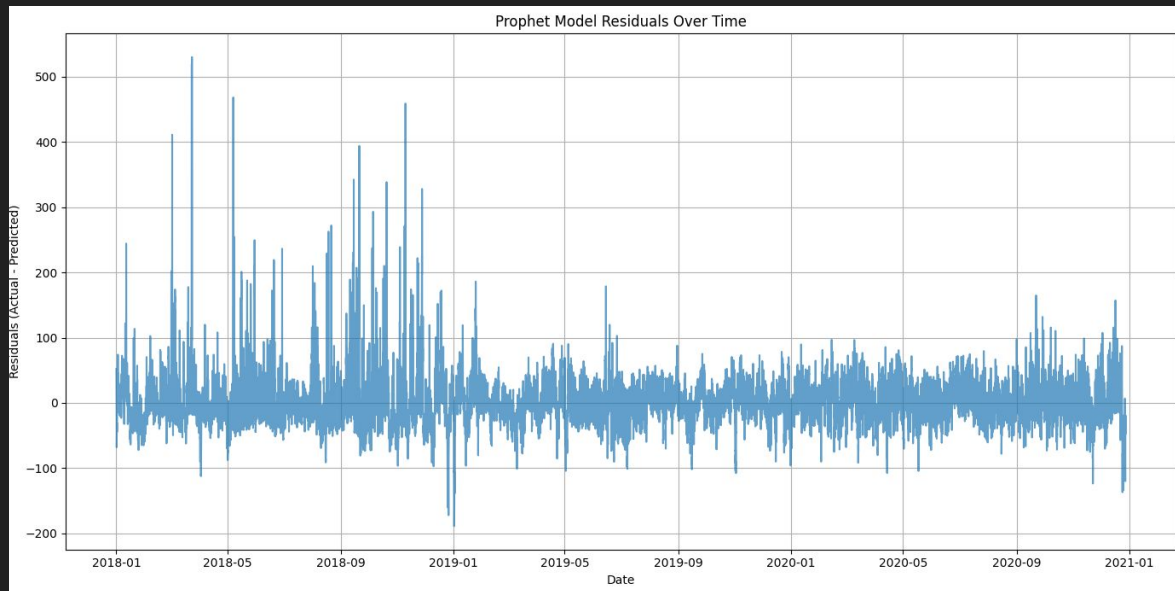
The error histogram is approximately centered around zero and close to a normal distribution, indicating unbiased predictions with most errors concentrated in a relatively narrow range



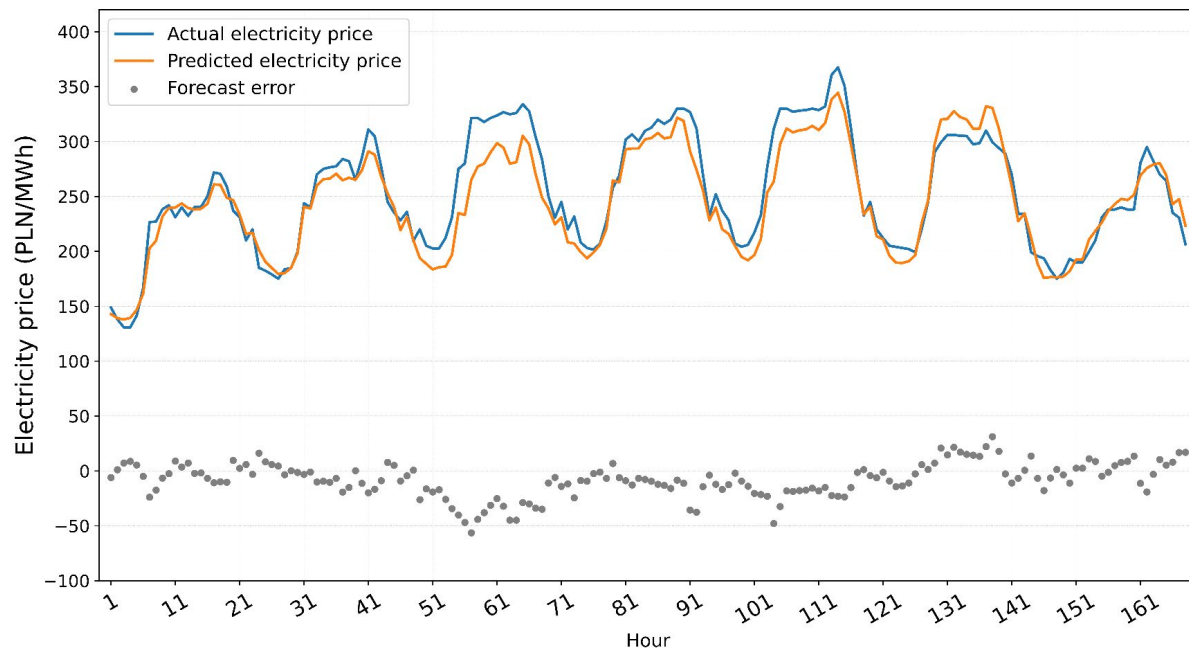
The residuals fluctuate around zero over time without a clear systematic pattern, suggesting that the MLP model does not exhibit strong temporal bias in its prediction errors



Together, these plots show that while the Prophet model captures major seasonal patterns, its residuals remain large and volatile over time, indicating limited accuracy in modeling short term price fluctuations



This plot shows that the model accurately captures the weekly cyclical pattern of electricity prices, with prediction errors remaining relatively small and stable across the entire week



This plot shows that the model closely tracks daily price patterns in December 2020, while forecast errors remain relatively small but increase during sudden price drops and peaks

